
```
!pip install xgboost
```

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.svm import SVR
from xgboost import XGBRegressor
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
```

```
uploaded = files.upload()
df = pd.read_excel(next(iter(uploaded)))
display(df.head())
```

```
features = [
    'Joint_Angle', 'Arm_Length', 'Payload', 'Speed', 'Torque',
    'Task_Complexity', 'Calibration_Frequency', 'Controller_Type',
    'Workspace_Temp', 'Cycle_Count'
]
target = 'Deviation'
```

```
X = df[features]
y = df[target]
X = pd.get_dummies(X, columns=['Controller_Type'], drop_first=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
models = {
    "Linear Regression": LinearRegression(),
    "Random Forest": RandomForestRegressor(random_state=42),
    "Decision Tree": DecisionTreeRegressor(random_state=42),
    "Support Vector Regressor": SVR(),
    "XGBoost": XGBRegressor(random_state=42, verbosity=0)
}
```

```
results = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
```

```

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
results[name] = {"MSE": mse, "R2 Score": r2}

print("\nModel Performance:")
for name, metrics in results.items():
    print(f"{name}:")
    print(f" - Mean Squared Error (MSE): {metrics['MSE']:.4f}")
    print(f" - R2 Score: {metrics['R2 Score']:.4f}")
    print()

model_names = list(results.keys())
r2_scores = [results[name]['R2 Score'] for name in model_names]
mses = [results[name]['MSE'] for name in model_names]

plt.figure(figsize=(8, 5))
plt.bar(model_names, r2_scores, color='mediumseagreen')
plt.ylabel("R2 Score")
plt.title("R2 Score Comparison Across Models")
plt.ylim(0, 1)
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
plt.plot(model_names, mses, marker='o', linestyle='-', color='dodgerblue')
plt.ylabel("Mean Squared Error")
plt.title("MSE Trend Across Models")
plt.xticks(rotation=45)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

total_r2 = sum(r2_scores)
r2_percentages = [score / total_r2 for score in r2_scores]

plt.figure(figsize=(7, 7))
plt.pie(r2_percentages, labels=model_names, autopct='%1.1f%%', startangle=140, colors=plt.cm.Set3.colors)
plt.title("Model Contribution Based on R2 Score")
plt.tight_layout()
plt.show()

```

➡ Requirement already satisfied: xgboost in /usr/local/lib/python3.11/dist-packages (2.1.4)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from xgboost) (2.0.2)
Requirement already satisfied: nvidia-nccl-cu12 in /usr/local/lib/python3.11/dist-packages (from xgboost) (2.21.5)
Requirement already satisfied: scipy in /usr/local/lib/python3.11/dist-packages (from xgboost) (1.15.2)

Choose Files robot_arm...n_data.xlsx

• **robot_arm_deviation_data.xlsx**(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 127259 bytes, last modified: 4/29/2025 - 100% done

Saving robot_arm_deviation_data.xlsx to robot_arm_deviation_data.xlsx

	Joint_Angle	Arm_Length	Payload	Speed	Torque	Task_Complexity	Calibration_Frequency	Controller_Type	Workspace_Temp	Cycle
0	67.417221	0.870266	2.690886	1.378136	59.339608	3	26.944393	MPC	34.851954	
1	171.128575	1.583802	2.545090	1.613695	81.516071	3	28.062745	MPC	28.180694	
2	131.758910	2.245892	9.071920	0.575889	77.215288	4	29.394106	RL	16.009111	
3	107.758527	1.964450	2.570507	1.287261	19.620491	3	26.056585	MPC	17.091257	
4	28.083355	2.113122	2.792302	1.186317	19.178700	5	5.218827	PID	32.620082	

Model Performance:

Linear Regression:

- Mean Squared Error (MSE): 0.9353
- R² Score: 0.9760

Random Forest:

- Mean Squared Error (MSE): 1.6117
- R² Score: 0.9587

Decision Tree:

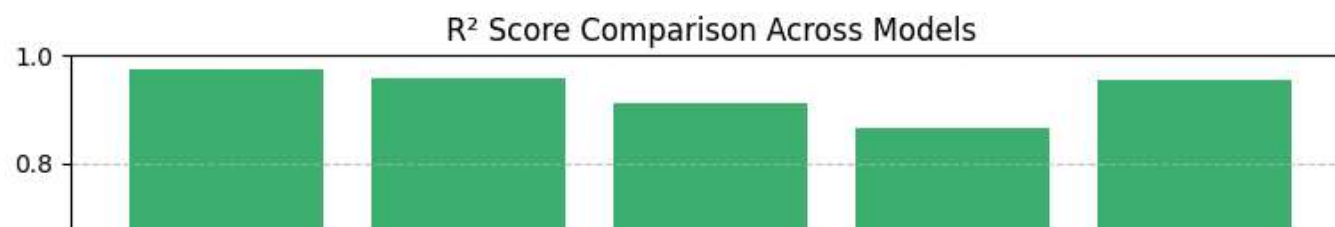
- Mean Squared Error (MSE): 3.4436
- R² Score: 0.9117

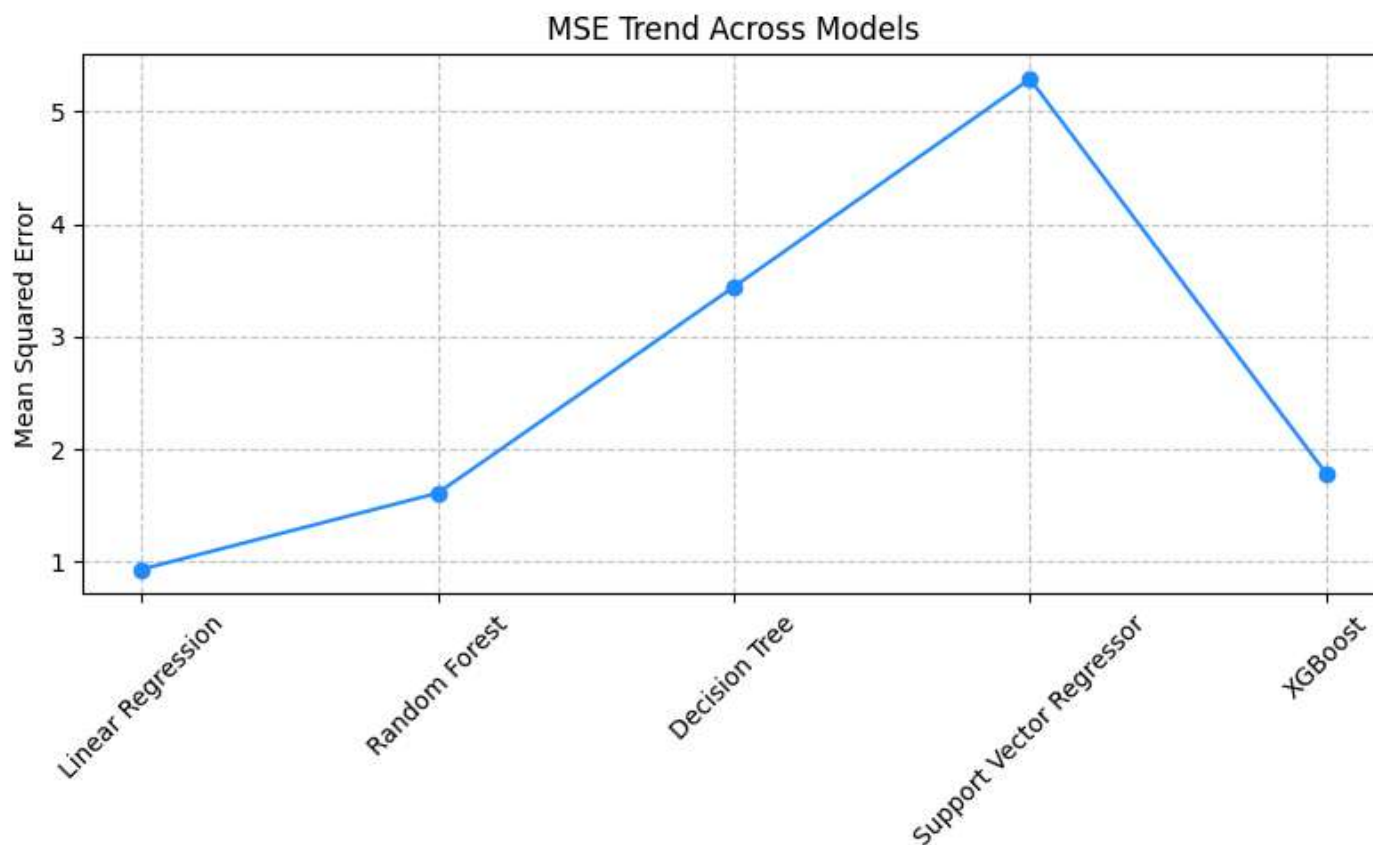
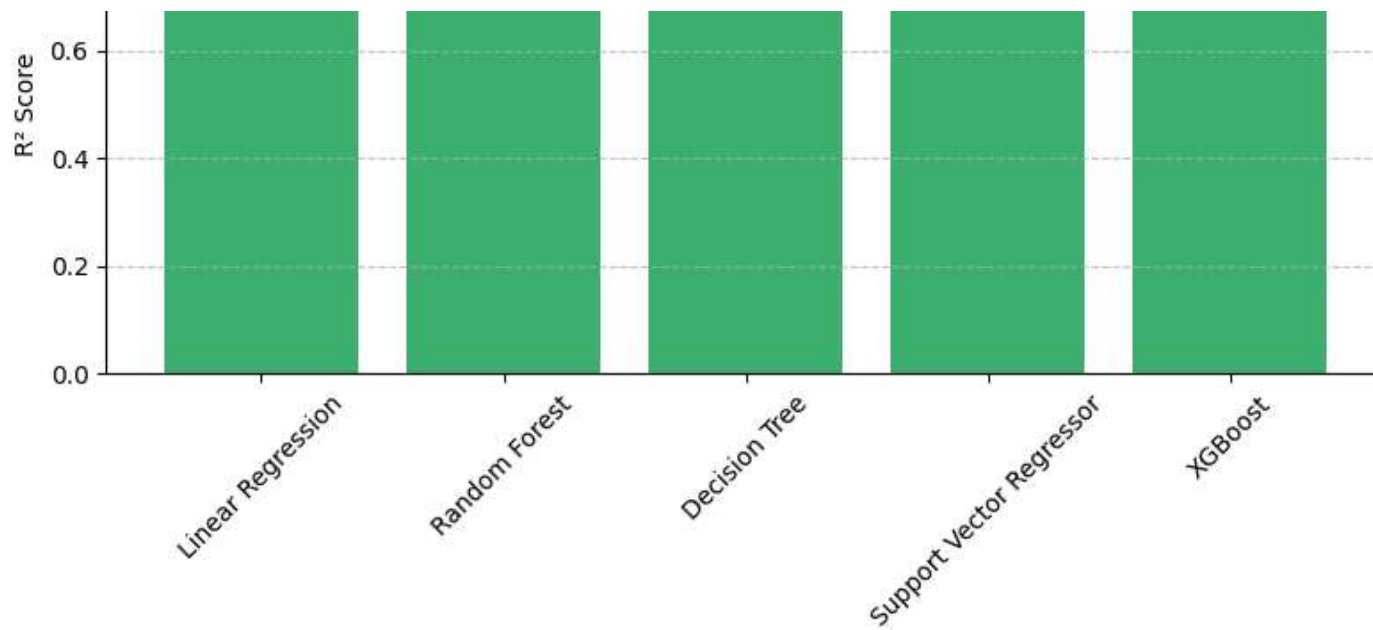
Support Vector Regressor:

- Mean Squared Error (MSE): 5.2883
- R² Score: 0.8645

XGBoost:

- Mean Squared Error (MSE): 1.7870
- R² Score: 0.9542





Model Contribution Based on R^2 Score

